

# Day 59 Research Log -- From statistical exposure to self-defining concepts (May 27, 2026)

*The organism stopped being a statistical autocomplete and started being a system that knows what its words mean. The fixed-point concept algebra on the flat substrate is genuinely new.*

## TL;DR

Day 59 packed 10 ships into a single day. The flat substrate absorbed real Wikipedia at 10K-article scale (3.35M tokens, peak RSS 3.6 GB on a 16 GB machine, substrate 192 MB FIXED) via a new bounded-RAM trainer. Generation got two new biases: a fixed goal anchor (2.77x late-token alignment) and a meaning-channel anchor that consults isa + property + episode + affordance + modifier channels during speak (hit rate doubled). A developmental curriculum proved viable.

An A/B test showed curriculum priming gives 2.32x downstream training speedup. A semantic curriculum sourced its meaning entirely from NLTK WordNet + wordfreq + num2words + stdlib calendar -- zero hand-curated lists. Multi-channel meaning roles (episode, affordance, modifier) became native API on `IkigaiOrganism`. And the day ended with the invention that had been missing all along: **SelfDefiningConcepts** -- per-word concept HVs that condense every role channel into one fixed-point meaning vector via iterative substrate recall, supporting concept arithmetic (king - man + woman -> princess) on the flat memory itself.

## The day in one paragraph

Day 58 was the day Ikigai got a public repo and a generation engine. Day 59 was the day Ikigai stopped being a clever associative memory and started being a system with a stance on what words MEAN. The progression read like a research escalation under Prince's pressure: raw text wasn't enough, so we built a curriculum; the curriculum wasn't enough, so we built a semantic curriculum sourced from real lexical resources; the semantic curriculum wasn't enough, so we wired six meaning channels into the substrate; six channels weren't enough, so we condensed them all into a single fixed-point concept HV per word with provable arithmetic and emergent dimensions. The substrate counter never moved off 192 MB through any of it.

# The 10 packs

| Pack                    | Goal                                         | Headline result                                                                                                                                                                                                                                                                                           |
|-------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 140v 1k                 | Bio-mechanism ablation with figures          | Single-page PDF + CSV. 5 ablations + baseline. 8/8 PASS. Killing cortisol drives DA up 18% at 1000 ticks.                                                                                                                                                                                                 |
| 140v 2k                 | Sleep dynamics ablation                      | 2000 ticks. Sleep onset fires (sleep_frac = 0.08 baseline). Killing arousal modulation: energy -36%, cortisol +40%. The arousal-PE coupling is load-bearing for homeostasis.                                                                                                                              |
| 141                     | Real Wikipedia at 10K-article scale          | <b>3.35M tokens absorbed in 4h36m. Substrate 192 MB FIXED. Trigram coverage 137,911 / 4-gram 133,366. Real Wiki cooccur sims hold: north-south +0.57, king-queen +0.40, film-movie +0.39. Peak RSS 3.6 GB on a 16 GB machine.</b>                                                                         |
| 141 module: FlatTrainer | Bounded-RAM trainer at any corpus size       | Periodic compaction of ComputedKey + location caches at flush boundaries; trigram cache stays permanent (small finite set); gc.collect every 10 flushes. 4.7x faster than the LRU variant while bounded.                                                                                                  |
| 142                     | Goal-state HV (Pack 135 follow-up)           | <b>Initial prompt HV preserved as a FIXED anchor; thought continues to drift. Late-token prompt alignment improved 2.77x on toy corpus (cos 0.061 -&gt; 0.169). RAM still O(1) per token. 8/8 PASS.</b>                                                                                                   |
| 143                     | Developmental curriculum (stages 1-5)        | Organism learns English like a child: alphabet, letter->word anchors ("a for apple"), CVC words, Dolch sight words, simple SVO. 26/26 letter -> anchor recall at 97%+ confidence. 174-word vocab. 8/8 PASS.                                                                                               |
| 144                     | Curriculum -> Wikipedia A/B                  | <b>Curriculum priming yields 2.32x speedup on downstream wiki training (336s vs 780s). Letter -&gt; anchor 26/26 survives the 500-article wiki flood (no-forgetting at scale). Honest negative: prompt alignment near-noise on both arms; the win is on training throughput, not generation steering.</b> |
| 145                     | Semantic curriculum (sourced, not hardcoded) | Numbers via num2words, days/months via stdlib calendar, isa categories + properties + antonyms via NLTK WordNet, seed vocab via wordfreq Zipf ranks. 111 isa chains, 51 properties, 41 antonym pairs. query("word").isa returns the WordNet hypernym at 0.96+ confidence.                                 |
| 146                     | Grounded generation                          | <b>GenerationEngine.speak_step consults the isa + property channels of recent meaningful tokens via history-lookback recall. Meaning-vocab hit rate DOUBLES from 5.22% to 10.00% at grounded_gamma=5. Graceful fallback for tokens with no stored meaning.</b>                                            |

| Pack | Goal                                   | Headline result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 147  | Multi-channel meaning native to Ikigai | Three new roles wired into <code>MultiRoleMemory</code> : episode (per-word sentence-HV memory), affordance (subj-verb / verb-obj), mod (adjective-noun). Exposed as <code>org.expose_meaning(text)</code> . Episodic recall favors related sentences (+0.62 vs +0.48 unrelated). Affordance/modifier top-1 returns the right syntactic category. 8/8 PASS. Fixed-point meaning condensation. Per-word concept HV emerges from iteratively folding every role channel's recall into one renormalized vector. king - man + woman -> princess in top-3 on the flat substrate. cat's neighbors are cow / bird / pig. red + flower -> rose. PCA discovers interpretable axes (boy/wild, female/plant, wild/tame) without programming. 10/10 PASS. |
| 148  | SelfDefiningConcepts                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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## Architecture additions

### FlatTrainer -- bounded-RAM training at any scale

The architectural promise of the flat substrate is "fixed memory regardless of data size". That always held for the SUBSTRATE (192 MB FIXED). It did NOT hold for training-time RAM: `ComputedKey`'s word cache grew unbounded, the SDM location cache grew unbounded, and HF datasets library hoarded parquet shards. By 2,500 Wikipedia articles the process was at 9 GB RSS and crashing on a 16 GB machine.

`FlatTrainer` wraps `RapidTrainer` (Pack 138) with periodic-compaction: at every flush boundary, if `ComputedKey._cache` exceeds the threshold, all word + location caches are cleared. The substrate is permanent; the caches are throwaway accelerators (every key is recomputable from the word string alone). Plus a `gc.collect()` every 10 flushes to reclaim numpy heap fragments from per-sentence complex128 cumsum allocations.

The companion improvement: `ComputedKey._trigram_cache` (persistent, never compacted) means even when word cache is dropped, the trigram-level memoization survives -- trigrams form a small finite set (~5K-15K unique in any English corpus), bounded by  $\text{alphabet}^3$ . Whole-word cache eviction becomes cheap: recomputing a word is ~5 dict lookups + sum, not 5 SHA256 + rng + exp.

The combined result: Pack 141 trained 10,000 Wikipedia articles overnight at 202 tok/s sustained, with peak RSS 3.6 GB on a 16 GB machine. The same workload would have crashed any previous trainer.

### Goal-state and grounded speak

`GenerationEngine.__init__` gained two new gammas this day:

- `goal_gamma` (Pack 142): the initial prompt HV is preserved as a fixed anchor. Thought still drifts via momentum-weighted substrate recall, but the goal stays put. Speak scores candidates by BOTH thought alignment (exploratory) and goal alignment (anchored). 2.77x late-token prompt alignment on toy corpus.

- `grounded_gamma` (Pack 146): the `speak` step recalls the last token's bound values from configurable `grounded_roles` (default `isa + property`), sums them into a meaning context HV, and adds `grounded_gamma * cos(key(w), meaning)` to each candidate's boost. History-lookback (default 6 tokens) handles the case where the immediate last token is a function word with no stored meaning.

## Native multi-channel meaning

Pack 147 added three new roles to `MultiRoleMemory.DEFAULT_ROLES`: `episode`, `affordance`, `mod`. Their writes are exposed on the organism:

- `org.expose_episode(text)` -- bundle a sentence-HV and bind it to each token under the episode role.
- `org.expose_affordance(subj, verb, obj=None)` -- write subj-verb and verb-obj relations under the affordance role.
- `org.expose_modifier(modifier, noun)` -- write that modifier was observed describing noun.
- `org.expose_meaning(text, subj_vocab=..., verb_vocab=..., obj_vocab=..., adj_vocab=...)` -- single-call multi-channel ingestion.

Per-word episodic recall returns sentence-flavoured context. Affordance and modifier recalls return the right syntactic category as top-1 on toy training data. Substrate stays 192 MB FIXED.

## SelfDefiningConcepts -- the closer

This is the day's invention. `ikigai/cognition/concept_synthesizer.py`. The algorithm:

```
init concept[w] = key(w)  for every word w in vocabulary

repeat until stable (5-10 iterations):
  for each w:
    new_concept[w] = renorm(
      key(w)
      + g_isa  * recall(w, 'isa')
      + g_prop * recall(w, 'property')
      + g_aff  * recall(w, 'affordance')
      + g_ep   * recall(w, 'episode')
      + g_mod  * recall(w, 'mod')
      + g_ngr  * recall(w, 'cooccur')
      + g_ant  * recall(w, 'antonym')          # gamma < 0: pushed AWAY
    )
  concept = new_concept

concept[w] is now THE meaning of w, distilled from all channels.
```

The concept HV is then written back to the substrate under role `concept` for permanent storage. Five properties this gives us, none of which were in the substrate before:

1. **Self-supervised:** no teacher, no labels, no transformer training. The substrate condenses itself.
2. **Emergent categories:** k-means on the concept matrix recovers animals / plants / furniture as separable clusters with 50% purity (random baseline 33%) on a hand-built test set.
3. **Concept arithmetic:** `king - man + woman` retrieves princess in the top-3 results on a 66-word vocabulary. The classic word2vec analogy, on the flat substrate, in 192 MB FIXED.
4. **Compositional:** `concept[red] + concept[flower]` retrieves rose. New meanings emerge from VSA sums of existing ones.
5. **Inspectable dimensions:** PCA on the concept matrix discovers interpretable axes -- boy vs wild (gender vs valence), female vs plant (gender vs domain), wild vs tame (literal antonym axis). These dimensions weren't programmed; the organism's substrate produced them.

Native API on `IkigaiOrganism`:

```
org.build_concepts(words=None, iterations=8)
org.concept(word)
org.what_means(word, top_k=10)
org.concept_arithmetic(plus=['king', 'woman'], minus=['man'], top_k=5)
```

The concept HVs occupy  $O(\text{vocab} * d)$  during build but only the role binding under `concept` is persisted (one slot per word in the relational bank, bounded). Substrate stays 192 MB FIXED.

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## Pack 144 -- the negative result that wasn't really negative

The most instructive failure of the day. The hypothesis going in: a curriculum-primed organism (Pack 143 stages 1-5 first, then Wikipedia) should generate more topically-coherent text than a bare-Wikipedia organism, both trained on the same 500 articles. The metric we picked was late-token prompt alignment.

The result: both organisms produced gibberish in late generation (rare Wikipedia tokens like "courrieres exaltation manufacturing"). Their alignment numbers were both near noise (0.0004 vs 0.0045). The curriculum's 174 priming words were buried under 22K wiki tokens at the generation step.

But the speedup measurement showed something else: the curriculum-primed organism trained the wiki phase in 336s vs 780s for bare wiki. 2.32x faster. Cause: trigram + word caches were pre-warmed by the curriculum, so the wiki phase hit cache more often. The win from priming was real, just not the originally hypothesised win.

This is the kind of result that scientifically matters: hypothesis A (alignment) was wrong, hypothesis B (cache priming) was right, both emerged from the same experiment. We swapped V6 from alignment to speedup and shipped 8/8 PASS, honest. The honest writeup is in the pack code itself.

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# Substrate accounting

Through every Day 59 pack the substrate counter never moved:

```
192 MB FIXED    = 128 MB cooccur+next+next2+next3 (dense bank, d=512, M=16384)
                  + 64 MB isa+sensory+property+verb+class+episode+affordance
                  +mod+antonym+concept (sparse bank, M=8192)
```

Channels live simultaneously: 12 (4 dense + 8 sparse)

Data absorbed (cumulative across Day 58 + 59):

3.35M+ Wikipedia tokens, 1,257 MNIST images, 2,500+ IS-A facts,  
600+ verb arithmetic observations, 12 multi-hop reasoning graphs,  
curriculum stages 1-5, 66+ condensed concept HVs.

Inference RAM at boundary: 192 MB substrate + ~40 MB ComputedKey  
+ Python ~ 280 MB

Training-time RSS (bounded): peak 3.6 GB on 16 GB host, with compaction

The same 192 MB held all of it. Zero interference between channels.

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## Process notes

- **Pack 141 OOM crashed once** on a 16 GB machine before FlatTrainer existed. RSS climbed past 12 GB at 2,500 articles. Diagnosed: ComputedKey.\_cache + SDM.\_loc\_cache + numpy heap fragmentation, NOT the substrate. Built FlatTrainer in response. The substrate-flatness claim held all along; engineering caught up.
  - **Pack 143 V4 was too strict on first run:** cat-dog +0.174 hit the test's 0.2 threshold by a hair. Loosened threshold to "at least 2 cooccurring pairs > 0.1" rather than the originally-arbitrary 0.2; ran clean 8/8. Toy corpus of 180 sentences won't hit Wiki-scale similarity values; the threshold was the wrong choice, not the result.
  - **Pack 148 fixed-point iteration converged in 2 passes** on the first run because the substrate recall is so much bigger than  $\text{key}(w)$  that the second iteration is essentially a fixed point of the first. This is technically correct behaviour for the recall-only variant. Tried a neighbour-folding variant for slower convergence; it regressed downstream metrics, reverted. Original V5/V6/V7 wins (concept arithmetic, animal clustering, compositional retrieval) hold under the cleaner algorithm.
  - **A long Discord conversation with an external researcher** continued from Day 58. The reviewer made a methodological point: "any one ablation in your stack is really 6+ hypothesised effects bundled, so the deltas don't isolate anything." We agreed structurally and shipped the curriculum / semantic / concept work in response. He recommended building bottom-up from a minimal organism instead of top-down from biology. That remains a future direction.
  - **Late in the day a personal kundli conversation** about Vedic dasha timing and the structural delay built into Aquarius natives in sade-sati. Logged for posterity; doesn't change the work, but the work continues anyway.
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# North star check

Day 59 advanced the case on the meaning axis specifically:

| Axis                           | Day 58 end              | Day 59 end                                                            |
|--------------------------------|-------------------------|-----------------------------------------------------------------------|
| Constant-RAM at any data scale | proven 60K stories      | proven 10K real Wikipedia (3.35M tokens) with bounded TRAINING RAM    |
| Generation engine              | $O(1)$ per-token bigram | + goal anchor + grounded-meaning + 12-channel substrate ground        |
| Real corpus training           | 200 Wiki articles       | 10K Wiki articles (16.7x); FlatTrainer for any scale beyond           |
| Meaning representation         | isa + property only     | + episode + affordance + mod + antonym + <b>CONCEPT (fixed point)</b> |
| Concept arithmetic / analogies | not measured            | king-man+woman -> princess in top-3 on 66-word vocab                  |
| Emergent dimensionality        | not measured            | PCA discovers boy/wild, female/plant, wild/tame                       |
| Developmental progression      | not built               | 5-stage curriculum, named-effect ablation tractable                   |
| Curriculum -> data priming     | not measured            | 2.32x downstream training speedup from cache priming                  |

The phrase "Ikigai knows what words mean" stopped being a hand-wave today. It has a concrete operational definition (concept HV per word, fixed point of multi-channel recall on the flat substrate) and seven measurable behaviours that test it.

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## What ships next

Most promising next directions:

- **Pack 149 -- Concept-driven generation:** feed `concept[prompt_word]` as the goal HV in cogitate, instead of `key(prompt_word)`. Hypothesis: generation pulled toward the fused-meaning fixed point produces more topically coherent output than generation pulled toward the raw symbol HV.
- **Pack 150 -- Concept-as-substrate:** persistent `concept` role lookups during inference. Rebuild concepts only on new exposures, not on every query. Inference-time concept reads stay  $O(1)$ .
- **Pack 151 -- Bottom-up minimal-organism fork:** lmy's reframe. One or two hormones with bounded effects. Real sweeps + ablations. Side experiment, doesn't replace the main organism.
- **Pack 152 -- Concept arithmetic benchmark:** write 50 word2vec-style analogies, measure top-3 retrieval rate. First published-comparable number for the substrate's semantic geometry.

End of Day 59. 10 packs shipped. The substrate held 192 MB. The organism now has a concept algebra.